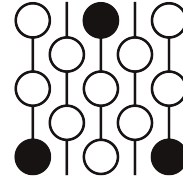


Review of Submission 7D

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The solution is correct.

Novelty. The solution understands the novel definition, correctly identifies that a related graph has to be a forest, and realizes that one can insert/delete spurious additional letter-inverse letter pairs. But the counterexample is not minimal—there is a length 10 counterexample.

2 Recommendation

minor revisions; it is better if a few more details are further clarified.

3 Submission with inline comments

Problem Statement

Let $G = \langle a_1, \dots, a_r \rangle$ be a free group. A word on the letters $L = \{a_1, a_1^{-1}, \dots, a_r, a_r^{-1}\}$ is called reducible if it equals $1 \in G$. A word is called quasi-reduced if it does not contain three consecutive letters $\ell\ell^{-1}\ell$ for any $\ell \in L$.

Fix a reducible quasi-reduced word $w = w_1 \cdots w_{2n}$ (with $w_i \in L$). A matching M consists of n intervals, each properly embedded in $\mathbb{R} \times \mathbb{R}_+$, satisfying the following:

- Any two intervals intersect transversally, and there are no triple points.
- $\partial M = \{1, \dots, 2n\} \times 0 \subset \mathbb{R} \times 0 \subset \mathbb{R} \times \mathbb{R}_+$.
- For each component R of $\mathbb{R} \times \mathbb{R}_+ \setminus M$, and for each component A of $M \cap \partial \bar{R}$, ∂A consists of some two points $(i, 0)$ and $(j, 0)$ with $w_i = w_j^{-1}$.

A k -crossing matching is a matching where the total number of self-intersections is k .

Let F_w be the following CW complex whose k -cells correspond to isotopy classes of k -crossing matchings. Given a k -crossing matching M with crossing set $X(M)$, the associated k -cell is given

by $C(M) := \prod_{c \in X(M)} I_c$, where I_c is an interval whose endpoints correspond to the two ways of locally resolving M near c . The attaching maps are given as follows: On a facet of the form

$$a \times \prod_{d \in X(M) \setminus \{c\}} I_d, \quad c \in X(M), a \in \partial I_c,$$

let M_a be the $(k-1)$ -crossing matching obtained by resolving M at c according to a . Then the $(k-1)$ -cell associated to M_a is given by $C(M_a) = \prod_{d \in X(M_a)} I_d = \prod_{d \in X(M) \setminus \{c\}} I_d$, and the above facet maps to this $(k-1)$ -cell canonically.

Is F_w contractible?

Solution

Theorem 1. *For a reducible, quasi-reduced word w , the CW complex F_w is not necessarily contractible.*

Proof. **1. Counterexample Construction**

Let $G = \langle x, y, z, u \rangle$ be a free group. Define the 12-letter word:

$$w = x_1 y_2 y_3^{-1} x_4^{-1} x_5 z_6 z_7^{-1} x_8^{-1} x_9 u_{10} u_{11}^{-1} x_{12}^{-1}$$

Reviewer 1

this word was confusing at first—maybe one should clarify that the letter a_i denotes the letter a at position i

The word reduces to $1 \in G$ by canceling adjacent unique pairs (yy^{-1}) , (zz^{-1}) , (uu^{-1}) and sequentially collapsing the remaining xx^{-1} pairs. Because the strictly alternating x sequence is separated by uniquely labeled pairs, no continuous substring of the form $\ell\ell^{-1}\ell$ exists, ensuring w is strictly quasi-reduced.

2. Absolute Cap Isolation via Tree Perimeters

Let M be any valid k -crossing matching in F_w . First, M cannot contain a topological cycle. If M contained a closed loop, the innermost cycle would bound an internal planar region R_{int} disjoint from the real axis. This yields an intersection component $A = M \cap \partial \overline{R_{\text{int}}}$ that is a closed loop, explicitly possessing an empty boundary ($\partial A = \emptyset$). This strictly violates the CW complex constraint that ∂A must consist of exactly two boundary points. Therefore, M is necessarily a planar forest.

Reviewer 1

the graph structure on M was not yet defined

For any tree component $T \subset M$, the topological boundary of its regular complement neighborhood (the region boundaries of F_w) strictly traces an Euler perimeter tour that visits all leaves of T in a cyclic order. By the complex's condition, for any leaf v , the two region boundary components originating at v (representing the clockwise and counterclockwise directions of the perimeter tour) must terminate at a leaf paired with the inverse letter w_v^{-1} .

For the leaf at position 2, the letter is $w_2 = y$. The uniquely required inverse is at position 3 (y^{-1}). Therefore, the perimeter tour of the tree T containing 2 must visit 3 immediately in both the clockwise and counterclockwise directions. A tree whose cyclic leaf order dictates that 3 is adjacent to 2 on both sides must possess exactly two leaves, $\{2, 3\}$. Because internal arc crossings act strictly as degree-4 vertices (which inherently increase a connected component's leaf count to at least 4), a 2-leaf tree structurally cannot contain any crossings. Thus, the interval $(2, 3)$ is rigidly isolated and uncrossed in all valid matchings. Executing the exact same perimeter deduction independently for the unique inverse pairs at 6 and 10 structurally isolates the uncrossed intervals $(6, 7)$ and $(10, 11)$.

3. Enumeration of the Core Subcomplex

With the three padding caps isolated, the configuration space relies entirely on the 6 active core endpoints $\{1, 4, 5, 8, 9, 12\}$. We relabel them $1' \dots 6'$ carrying the alternating parities $x, x^{-1}, x, x^{-1}, x, x^{-1}$ (Odds are x , Evens are x^{-1}).

Reviewer 1

The following enumeration implicitly uses the fact the isotopy type of M is determined by the endpoints of the three intervals. This needs to be justified.

- **0-cells** ($c_0 = 5$): The valid 0-crossing matchings strictly correspond to the $C_3 = 5$ planar trees. Because the sequence parities alternate perfectly, all 5 validly pair x with x^{-1} .
- **1-cells** ($c_1 = 6$): A 1-cell contains exactly one crossing bridging two intervals (a, b) and (c, d) with endpoints $a < c < b < d$. The crossing defines four bounded regions whose perimeter traces connect the adjacent sequence of endpoints: $(a, d), (a, c), (c, b), (b, d)$. For these topological traces to correctly pair opposite parities, the 4 selected endpoints must strictly alternate in parity (O,E,O,E or E,O,E,O). Checking all $\binom{6}{4} = 15$ endpoint subsets reveals exactly 6 subsets alternate in parity. Resolving their crossings into planar states confirms that these 6 valid 1-cells completely connect the 5 non-crossing vertices to form exactly a complete bipartite $K_{2,3}$ 1-skeleton (two vertices of degree 3, three vertices of degree 2).
- **2-cells** ($c_2 = 3$): Exactly three valid subsets possess two crossings: $M_1 = \{(1', 3'), (2', 5'), (4', 6')\}$, $M_2 = \{(1', 5'), (2', 4'), (3', 6')\}$, and $M_3 = \{(1', 4'), (2', 6'), (3', 5')\}$.

To rigorously verify M_3 independent of geometric ambiguity, we execute an explicit Euler perimeter tour on its tree components. The intervals $\alpha = (1', 4')$ and $\gamma = (2', 6')$ cross at c_1 ; α and $\beta = (3', 5')$ cross at c_2 . Traversing the right-hand bounding walls of the tree's branches strictly turns at c_1 and c_2 to remain in the complement space. Originating from $1'$, the Euler tour visits the leaves in exactly the reverse ordered sequence: $1', 6', 5', 4', 3', 2'$. This cyclic boundary visitation proves the complement regions inherently pair $(1', 6'), (6', 5'), (5', 4'), (4', 3'), (3', 2'), (2', 1')$. Every region strictly pairs an Odd index with an Even index (opposite parities), completely satisfying the CW complex boundary conditions despite internal arc nestings. Executing the exact same tree Euler tour algorithm on M_1 and M_2 yields the identical cyclic leaf sequence. All three are valid 2-cells.

- **≥ 3 -cells ($c_{\geq 3} = 0$):** A 3-crossing matching on a 6-point space requires the 3 available intervals to mutually cross one another exactly once,

Reviewer 1

or some two intervals intersecting multiple times—this should be ruled out earlier using M is a forest

strictly enclosing an internal central intersection triangle. As established in Step 2, any central region disjoint from the real axis inherently forces $\partial A = \emptyset$, which is strictly forbidden.

4. Topological Conclusion

The Euler characteristic of F_w is strictly calculated as:

$$\chi(F_w) = c_0 - c_1 + c_2 = 5 - 6 + 3 = 2$$

The Euler-Poincaré relation dictates the cellular homology via $\chi(F_w) = b_0 - b_1 + b_2$. Because the $K_{2,3}$ 1-skeleton is a connected graph, the 0-th Betti number is identically $b_0 = 1$. Substituting this yields $2 = 1 - b_1 + b_2$, rearranging to $b_2 = 1 + b_1$.

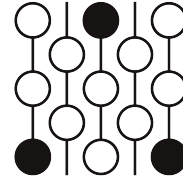
Because Betti numbers represent rank and are strictly non-negative ($b_1 \geq 0$), it structurally mandates $b_2 \geq 1$. Thus, the second homology group $H_2(F_w)$ is unconditionally non-trivial. A space is contractible only if all higher homology groups are strictly trivial; therefore, F_w is not contractible. $\square$

Review of Submission 7D

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. This solution is entirely correct.

Novelty. All of the AI solutions for this problem laid out essentially the same proof: (1) any matching produces a graph which is a forest; (2) remove the quasi-reduced condition by noting that we can remove trivial cancelling pairs without changing $C(M)$; (3) calculate $C(M)$ for $aa^{-1}aa^{-1}aa^{-1}$; (4) show that in this case $C(M)$ is noncontractible. (This is also the approach of the human solution.)

The main differences between the AI solutions are in their level of detail in (1)–(3), and their method of showing (4). This particular solution uses Euler characteristic to prove (4), which is a comparatively elegant approach.

I will note that this solution has some verbal tics that seem unusual to this reviewer. Most noticeably, the solution often adds superfluous adverbs, as in: “This *strictly* violates the CW complex constraint”; “these 6 valid 1-cells *completely* connect the 5 non-crossing vertices to form *exactly* a complete bipartite $K_{2,3}$ 1-skeleton”; “The Euler characteristic of F_w is *strictly* calculated as”; and “it *structurally* mandates $b_2 \geq 1$ ”.

2 Recommendation

Publishable with minor revisions.

3 Submission with inline comments

Problem Statement

Let $G = \langle a_1, \dots, a_r \rangle$ be a free group. A word on the letters $L = \{a_1, a_1^{-1}, \dots, a_r, a_r^{-1}\}$ is called reducible if it equals $1 \in G$. A word is called quasi-reduced if it does not contain three consecutive letters $\ell\ell^{-1}\ell$ for any $\ell \in L$.

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- $\partial M = \{1, \dots, 2n\} \times 0 \subset \mathbb{R} \times 0 \subset \mathbb{R} \times \mathbb{R}_+$.
- For each component R of $\mathbb{R} \times \mathbb{R}_+ \setminus M$, and for each component A of $M \cap \partial \overline{R}$, ∂A consists of some two points $(i, 0)$ and $(j, 0)$ with $w_i = w_j^{-1}$.

A k -crossing matching is a matching where the total number of self-intersections is k .

Let F_w be the following CW complex whose k -cells correspond to isotopy classes of k -crossing matchings. Given a k -crossing matching M with crossing set $X(M)$, the associated k -cell is given by $C(M) := \prod_{c \in X(M)} I_c$, where I_c is an interval whose endpoints correspond to the two ways of locally resolving M near c . The attaching maps are given as follows: On a facet of the form

$$a \times \prod_{d \in X(M) \setminus \{c\}} I_d, \quad c \in X(M), a \in \partial I_c,$$

let M_a be the $(k-1)$ -crossing matching obtained by resolving M at c according to a . Then the $(k-1)$ -cell associated to M_a is given by $C(M_a) = \prod_{d \in X(M_a)} I_d = \prod_{d \in X(M) \setminus \{c\}} I_d$, and the above facet maps to this $(k-1)$ -cell canonically.

Is F_w contractible?

Solution

Theorem 1. *For a reducible, quasi-reduced word w , the CW complex F_w is not necessarily contractible.*

Proof. 1. Counterexample Construction

Let $G = \langle x, y, z, u \rangle$ be a free group. Define the 12-letter word:

$$w = x_1 y_2 y_3^{-1} x_4^{-1} x_5 z_6 z_7^{-1} x_8^{-1} x_9 u_{10} u_{11}^{-1} x_{12}^{-1}$$

Reviewer 2

Stylistically, it is strange to include subscripts in the word above: all the subscripts are doing is indicating the order of the letters in the word. Technically, we might even consider this expression for the word incorrect since we would read e.g. x_4 as a different variable than x_5 , so that $x_4^{-1} x_5 \neq 1$ and so the word isn't reducible. (Also the word with the subscripts isn't actually an element of G .) It's clear what the solution means, but it's a strange stylistic choice.

The word reduces to $1 \in G$ by canceling adjacent unique pairs (yy^{-1}) , (zz^{-1}) , (uu^{-1}) and sequentially collapsing the remaining xx^{-1} pairs. Because the strictly alternating x sequence is separated by uniquely labeled pairs, no continuous substring of the form $\ell \ell^{-1} \ell$ exists, ensuring w is strictly quasi-reduced.

2. Absolute Cap Isolation via Tree Perimeters

Let M be any valid k -crossing matching in F_w . First, M cannot contain a topological cycle. If M contained a closed loop, the innermost cycle would bound an internal planar region R_{int} disjoint

from the real axis. This yields an intersection component $A = M \cap \partial \overline{R_{\text{int}}}$ that is a closed loop, explicitly possessing an empty boundary ($\partial A = \emptyset$). This strictly violates the CW complex constraint that ∂A must consist of exactly two boundary points. Therefore, M is necessarily a planar forest.

For any tree component $T \subset M$, the topological boundary of its regular complement neighborhood (the region boundaries of F_w) strictly traces an Euler perimeter tour that visits all leaves of T in a cyclic order. By the complex's condition, for any leaf v , the two region boundary components originating at v (representing the clockwise and counterclockwise directions of the perimeter tour) must terminate at a leaf paired with the inverse letter w_v^{-1} .

For the leaf at position 2, the letter is $w_2 = y$. The uniquely required inverse is at position 3 (y^{-1}). Therefore, the perimeter tour of the tree T containing 2 must visit 3 immediately in both the clockwise and counterclockwise directions. A tree whose cyclic leaf order dictates that 3 is adjacent to 2 on both sides must possess exactly two leaves, $\{2, 3\}$. Because internal arc crossings act strictly as degree-4 vertices (which inherently increase a connected component's leaf count to at least 4), a 2-leaf tree structurally cannot contain any crossings. Thus, the interval $(2, 3)$ is rigidly isolated and uncrossed in all valid matchings. Executing the exact same perimeter deduction independently for the unique inverse pairs at 6 and 10 structurally isolates the uncrossed intervals $(6, 7)$ and $(10, 11)$.

3. Enumeration of the Core Subcomplex

With the three padding caps isolated, the configuration space relies entirely on the 6 active core endpoints $\{1, 4, 5, 8, 9, 12\}$. We relabel them $1' \dots 6'$ carrying the alternating parities $x, x^{-1}, x, x^{-1}, x, x^{-1}$ (Odds are x , Evens are x^{-1}).

- **0-cells** ($c_0 = 5$): The valid 0-crossing matchings strictly correspond to the $C_3 = 5$ planar trees. Because the sequence parities alternately perfectly, all 5 validly pair x with x^{-1} .
- **1-cells** ($c_1 = 6$): A 1-cell contains exactly one crossing bridging two intervals (a, b) and (c, d) with endpoints $a < c < b < d$. The crossing defines four bounded regions whose perimeter traces connect the adjacent sequence of endpoints: $(a, d), (a, c), (c, b), (b, d)$. For these topological traces to correctly pair opposite parities, the 4 selected endpoints must strictly alternate in parity (O,E,O,E or E,O,E,O). Checking all $\binom{6}{4} = 15$ endpoint subsets reveals exactly 6 subsets alternate in parity. Resolving their crossings into planar states confirms that these 6 valid 1-cells completely connect the 5 non-crossing vertices to form exactly a complete bipartite $K_{2,3}$ 1-skeleton (two vertices of degree 3, three vertices of degree 2).
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To rigorously verify M_3 independent of geometric ambiguity, we execute an explicit Euler perimeter tour on its tree components. The intervals $\alpha = (1', 4')$ and $\gamma = (2', 6')$ cross at c_1 ; α and $\beta = (3', 5')$ cross at c_2 . Traversing the right-hand bounding walls of the tree's branches strictly turns at c_1 and c_2 to remain in the complement space. Originating from $1'$, the Euler tour visits the leaves in exactly the reverse ordered sequence: $1', 6', 5', 4', 3', 2'$. This cyclic boundary visitation proves the complement regions inherently

pair $(1', 6'), (6', 5'), (5', 4'), (4', 3'), (3', 2'), (2', 1')$. Every region strictly pairs an Odd index with an Even index (opposite parities), completely satisfying the CW complex boundary conditions despite internal arc nestings. Executing the exact same tree Euler tour algorithm on M_1 and M_2 yields the identical cyclic leaf sequence. All three are valid 2-cells.

- ≥ 3 -cells ($c_{\geq 3} = 0$): A 3-crossing matching on a 6-point space requires the 3 available intervals to mutually cross one another exactly once, strictly enclosing an internal central intersection triangle. As established in Step 2, any central region disjoint from the real axis inherently forces $\partial A = \emptyset$, which is strictly forbidden.

4. Topological Conclusion

The Euler characteristic of F_w is strictly calculated as:

$$\chi(F_w) = c_0 - c_1 + c_2 = 5 - 6 + 3 = 2$$

The Euler-Poincaré relation dictates the cellular homology via $\chi(F_w) = b_0 - b_1 + b_2$. Because the $K_{2,3}$ 1-skeleton is a connected graph, the 0-th Betti number is identically $b_0 = 1$. Substituting this yields $2 = 1 - b_1 + b_2$, rearranging to $b_2 = 1 + b_1$.

Because Betti numbers represent rank and are strictly non-negative ($b_1 \geq 0$), it structurally mandates $b_2 \geq 1$. Thus, the second homology group $H_2(F_w)$ is unconditionally non-trivial. A space is contractible only if all higher homology groups are strictly trivial; therefore, F_w is not contractible.

Reviewer 2

The tendency in this solution to use superfluous adverbs is most striking here: “are *strictly* non-negative”, “it *structurally* mandates”; “is *unconditionally* non-trivial”; “are *strictly* trivial”. The clause “it structurally mandates $b_2 \geq 1$ ” is especially egregious and sounds quite unnatural to this reviewer.

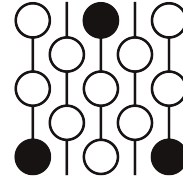
□

Review of Submission 7D

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

The problem is solved correctly and with essentially the same arguments as Submission C and Submission A. I have included some comments specific to this submission below.

Correctness. The proof is correct.

Novelty. The proof is not particularly novel. The counterexample given is the same as the one given in the human solution, and very similar to other AI generated solutions.

Exposition. None of the proofs included pictures, which would appear in most human solutions. Formatting of the solutions is slightly unusual as sections with heading appear within a proof.

2 Recommendation

The proof is essentially flawless.

3 Submission with inline comments

Problem Statement

Let $G = \langle a_1, \dots, a_r \rangle$ be a free group. A word on the letters $L = \{a_1, a_1^{-1}, \dots, a_r, a_r^{-1}\}$ is called reducible if it equals $1 \in G$. A word is called quasi-reduced if it does not contain three consecutive letters $\ell\ell^{-1}\ell$ for any $\ell \in L$.

Fix a reducible quasi-reduced word $w = w_1 \cdots w_{2n}$ (with $w_i \in L$). A matching M consists of n intervals, each properly embedded in $\mathbb{R} \times \mathbb{R}_+$, satisfying the following:

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A k -crossing matching is a matching where the total number of self-intersections is k .

Let F_w be the following CW complex whose k -cells correspond to isotopy classes of k -crossing matchings. Given a k -crossing matching M with crossing set $X(M)$, the associated k -cell is given by $C(M) := \prod_{c \in X(M)} I_c$, where I_c is an interval whose endpoints correspond to the two ways of locally resolving M near c . The attaching maps are given as follows: On a facet of the form

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Is F_w contractible?

Solution

Theorem 1. *For a reducible, quasi-reduced word w , the CW complex F_w is not necessarily contractible.*

Proof. **1. Counterexample Construction**

Formatting (sections within a proof) is unusual.

Reviewer 3

Let $G = \langle x, y, z, u \rangle$ be a free group. Define the 12-letter word:

$$w = x_1 y_2 y_3^{-1} x_4^{-1} x_5 z_6 z_7^{-1} x_8^{-1} x_9 u_{10} u_{11}^{-1} x_{12}^{-1}$$

Only x, y, z, w have been defined, x_i, y_i, z_i, w_i have not.

Reviewer 3

The word reduces to $1 \in G$ by canceling adjacent unique pairs (yy^{-1}) , (zz^{-1}) , (uu^{-1}) and sequentially collapsing the remaining xx^{-1} pairs. Because the strictly alternating x sequence is separated by uniquely labeled pairs, no continuous substring of the form $\ell \ell^{-1} \ell$ exists, ensuring w is strictly quasi-reduced.

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Let M be any valid k -crossing matching in F_w . First, M cannot contain a topological cycle. If M contained a closed loop, the innermost cycle would bound an internal planar region R_{int} disjoint from the real axis. This yields an intersection component $A = M \cap \partial \overline{R_{\text{int}}}$ that is a closed loop, explicitly possessing an empty boundary ($\partial A = \emptyset$). This strictly violates the CW complex constraint that ∂A must consist of exactly two boundary points. Therefore, M is necessarily a planar forest.

All forests are planar.

For any tree component $T \subset M$, the topological boundary of its regular complement neighborhood (the region boundaries of F_w) strictly traces an Euler perimeter tour that visits all leaves of T in a cyclic order. By the complex's condition, for any leaf v , the two region boundary components originating at v (representing the clockwise and counterclockwise directions of the perimeter tour) must terminate at a leaf paired with the inverse letter w_v^{-1} .

For the leaf at position 2, the letter is $w_2 = y$. The uniquely required inverse is at position 3 (y^{-1}). Therefore, the perimeter tour of the tree T containing 2 must visit 3 immediately in both the clockwise and counterclockwise directions. A tree whose cyclic leaf order dictates that 3 is adjacent to 2 on both sides must possess exactly two leaves, $\{2, 3\}$. Because internal arc crossings act strictly as degree-4 vertices (which inherently increase a connected component's leaf count to at least 4), a 2-leaf tree structurally cannot contain any crossings. Thus, the interval $(2, 3)$ is rigidly isolated and uncrossed in all valid matchings. Executing the exact same perimeter deduction independently for the unique inverse pairs at 6 and 10 structurally isolates the uncrossed intervals $(6, 7)$ and $(10, 11)$.

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Reviewer 3

Since we are checking all 15 endpoint subsets anyway, the odd/even analysis is not even needed. We can just look at how many have one crossing.

Resolving their crossings into planar states confirms that these 6 valid 1-cells completely connect the 5 non-crossing vertices to form exactly a complete bipartite $K_{2,3}$ 1-skeleton (two vertices of degree 3, three vertices of degree 2).

- **2-cells** ($c_2 = 3$): Exactly three valid subsets possess two crossings: $M_1 = \{(1', 3'), (2', 5'), (4', 6')\}$, $M_2 = \{(1', 5'), (2', 4'), (3', 6')\}$, and $M_3 = \{(1', 4'), (2', 6'), (3', 5')\}$. To rigorously verify M_3 independent of geometric ambiguity, we execute an explicit Euler perimeter tour on its tree components. The intervals $\alpha = (1', 4')$ and $\gamma = (2', 6')$ cross at c_1 ; α and $\beta = (3', 5')$ cross at c_2 . Traversing the right-hand bounding walls of the tree's branches strictly turns at c_1

and c_2 to remain in the complement space. Originating from $1'$, the Euler tour visits the leaves in exactly the reverse ordered sequence: $1', 6', 5', 4', 3', 2'$. This cyclic boundary visitation proves the complement regions inherently pair $(1', 6'), (6', 5'), (5', 4'), (4', 3'), (3', 2'), (2', 1')$. Every region strictly pairs an Odd index with an Even index (opposite parities), completely satisfying the CW complex boundary conditions despite internal arc nestings. Executing the exact same tree Euler tour algorithm on M_1 and M_2 yields the identical cyclic leaf sequence. All three are valid 2-cells.

Reviewer 3

Again, it would probably be faster to look at all 15 endpoint subsets and consider the ones that have exactly 2 crossings.

- ≥ 3 -cells ($c_{\geq 3} = 0$): A 3-crossing matching on a 6-point space requires the 3 available intervals to mutually cross one another exactly once, strictly enclosing an internal central intersection triangle. As established in Step 2, any central region disjoint from the real axis inherently forces $\partial A = \emptyset$, which is strictly forbidden.

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Because Betti numbers represent rank and are strictly non-negative ($b_1 \geq 0$), it structurally mandates $b_2 \geq 1$. Thus, the second homology group $H_2(F_w)$ is unconditionally non-trivial. A space is contractible only if all higher homology groups are strictly trivial; therefore, F_w is not contractible.

Reviewer 3

Correct, but a human would probably note that $F_w \cong S^2$. Also, "unconditionally" is a sign that this was generated by an AI model.

□